Software Test Plan (STP)

Remote Lab Reservation System (RLRS) — IEEE 829 Standard

Project	Remote Lab Reservation System (RLRS)
Version	1.1
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Date	11 November 2026
Status	Updated to match the implemented system (v1.1)

Revision History

Version	Date	Author	Change summary	Approval
1.0	30 October 2026	Team 11	Initial project-specific IEEE baseline specification.	Approved
1.1	11 November 2026	Team 11	Updated to v1.1: Dedicated Section 5.1 to Security Validation, filled Team 11 roles and schedule, mapped all 33 test cases to automated pytest/Playwright tests, and recorded 71/71 passing test results.	Approved

Approvals

Role	Name	Signature / SRN / Email	Date
Project team representative	Mahipal Singh	PES1UG24AM153 (Team Leader)	11 November 2026
Scheduling & backend lead	Lakshya Budhauliya	PES1UG24AM147	11 November 2026
Calendar UI, auth & email lead	Deepesh M	PES1UG24AM150 (mdeepesh2006@gmail.com)	11 November 2026
QA & documentation member	Venkat Kushal	PES1UG24AM124	11 November 2026
Course instructor / TA	Kshitij Shetty	SE Teaching Assistant, Dept. of CSE(AIML)	Pending Review

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1. Introduction

1.1 Purpose

This Software Test Plan (STP) defines the scope, security validation strategy, test levels, environment, team responsibilities, traceability, and execution results for verifying the Remote Lab Reservation System (RLRS).

1.2 Scope

Testing covers authentication and role authorization, lab and equipment management, operating hours and slot grid generation, reservation creation, conflict detection, booking rules, restricted-equipment approval workflow, cancellation cut-off, maintenance blocks, confirmation and reminder emails with retry, holder-only remote-access visibility, attendance marking, utilisation reporting, audit logging, and security/concurrency non-functional requirements.

2. Test Items

- Pure scheduling & domain engine: rlrs_core/conflict.py, rlrs_core/rules.py, rlrs_core/reminders.py, rlrs_core/maintenance.py, rlrs_core/attendance.py, rlrs_core/audit.py, rlrs_core/report.py
- Service & persistence layer: rlrs_app/service.py, rlrs_app/db.py, rlrs_app/security.py, rlrs_app/mailer.py, rlrs_app/seed.py
- HTTP REST API & background worker: rlrs_app/server.py
- Browser week-calendar frontend: rlrs_app/static/index.html, rlrs_app/static/app.js, rlrs_app/static/style.css

3. Features to be Tested

Requirement / Feature	Test Focus
RLRS-F-001 to RLRS-F-002	Valid/invalid login, session expiry, logout, role restrictions, direct API calls by unauthorized roles.

Requirement / Feature	Test Focus
RLRS-F-003 to RLRS-F-005	Equipment create/edit/deactivate, lab operating hours, slot length changes, equipment search/filtering.
RLRS-F-006	Weekly calendar grid slot states (free, own, booked, maintenance, past) and date-range validation.
RLRS-F-007 to RLRS-F-008	Booking free slots; partial overlap, full containment, identical slot, and back-to-back boundary cases.
RLRS-F-009	Every booking rule: past start, outside hours, multi-day, off-grid, advance window, max duration, max active bookings, same-user overlap.
RLRS-F-010 to RLRS-F-012, F-018	Pending approval & rejection with reason, cross-lab in-charge denial, My Reservations list, 60-min cancellation cut-off, attendance marking.
RLRS-F-013	Maintenance block creation, automatic cancellation of overlapping reservations, holder notification, and blocking new bookings.
RLRS-F-014 to RLRS-F-016	Confirmation/cancellation/approval emails, 2-hour reminder scheduling, single delivery across restarts, retry with backoff on SMTP failure.
RLRS-F-017 / RLRS-SR-005	Record-level access: remote-access instructions visible only to Confirmed reservation holders and lab staff; students cannot cancel others' bookings.
RLRS-F-019, RLRS-F-021	Utilisation and no-show report reconciliation; append-only audit log with secret redaction.
RLRS-NF-002 to RLRS-NF-008	10-thread parallel double-booking prevention, restart durability, PBKDF2 credential security, SQLi/path-traversal immunity, cross-OS CI portability.

4. Features Not to be Tested

- Physical lab hardware operation or live SSH/VNC remote-equipment sessions (RLRS manages scheduling and access instructions only).
- External institutional SMTP relay deliverability and spam filtering (tests verify hand-off using FileTransport and simulated SMTP failure stubs).
- 20,000-reservation bulk performance benchmark (RLRS-NF-001; documented as not run in this prototype release).
- University SSO, iCal export, and SMS notifications (outside project scope).

5. Test Approach / Strategy

5.1 Security Validation

Security validation is a mandatory pillar of the RLRS test strategy and verifies every control defined in SRS Section 6 (RLRS-SR-001 through RLRS-SR-007 and RLRS-NF-005) at both the service layer and over real HTTP:

- Authentication & Session Security (TC-AUTH-01, TC-SEC-01): Verify unauthenticated requests to protected endpoints return 401; verify session cookies include HttpOnly and SameSite=Lax flags; verify expired sessions are rejected.
- Server-Side Role-Based Authorization (TC-AUTH-02, RLRS-SR-002): Verify Student accounts calling staff endpoints (POST /api/equipment, POST /api/maintenance, POST /api/reservations/{id}/approve, GET /api/audit, GET /api/reports/utilisation) receive 403 Forbidden.
- Record-Level Ownership & Cross-Lab Isolation (TC-SEC-02, RLRS-SR-005, RLRS-F-017): Verify Student B cannot cancel Student A's reservation (403); verify equipment remote_info is hidden from students unless they hold a Confirmed reservation for that equipment; verify a Lab In-charge assigned to Lab 1 gets 403 when attempting to approve/reject bookings in Lab 2.

- Credential Hashing & Audit Redaction (TC-SEC-01, TC-AUDIT-01, RLRS-SR-003, RLRS-SR-007): Verify user passwords are stored exclusively as salted PBKDF2-HMAC-SHA256 hashes (120,000 iterations) and that plaintext passwords, hashes, and tokens never appear in /api/me responses or audit logs.
- Input Validation, SQL Injection & Path Traversal Defence (TC-SEC-03, TC-SEC-04, TC-ERR-01, RLRS-SR-004, RLRS-F-022): Verify SQL injection payloads in login and search parameters are inert (parameterized SQL); verify URL path traversal attempts (/../service.py, /%2e%2e/service.py) return 404; verify malformed JSON or invalid IDs return structured 400 JSON without stack traces.

5.2 Test Levels

- Unit Testing (37 tests in tests/unit/): Validate pure scheduling logic (interval overlap, all 7 booking rules, reminder state machine, maintenance cancellation, attendance transitions, audit redaction, utilisation math) with zero I/O.
- Integration & Concurrency Testing (24 tests in tests/integration/test_service.py): Validate service operations against real SQLite databases, including UTC storage conversion, 10-thread parallel booking serialization, and email/reminder worker retries.
- HTTP API Contract & Security Testing (10 tests in tests/integration/test_api.py): Validate live ThreadingHTTPServer endpoints, status codes, cookies, SQLi, and path traversal protections.
- End-to-End Browser UI Testing (9 steps in tests/ui/ui_smoke.py): Validate user flows in headless Chromium via Playwright.

5.3 Test Types, 5.4 Entry Criteria & 5.5 Exit Criteria

- Test Types: Functional, boundary-value (half-open intervals, operating-hour edges, 60-min cutoff), concurrency (10-thread Barrier), time-travel (controllable clock), API contract, and security testing.
- Entry Criteria: Runnable stdlib server and SQLite schema available; deterministic seed fixtures loaded.
- Exit Criteria: 100% of automated unit, integration, and API tests pass in CI across Ubuntu, Windows, and macOS; zero double-booking or authorization defects remain; all deviations documented.

6. Test Environment

Area	Configuration (as executed in v1.1)
Hardware / OS	GitHub Actions runners (ubuntu-latest, windows-latest, macos-latest) + local macOS (Apple Silicon, Darwin) and Windows laptops.
Software	Python 3.12 (CI) / Python 3.13 & 3.14 (local); Python standard library (http.server, sqlite3, threading, hashlib); SQLite 3.
Browser	Chromium (automated via Playwright in tests/ui/ui_smoke.py).
Tools	pytest 9.x, controllable clock injection, FileTransport mail catcher, stub failing SMTP transport, GitHub Actions CI (.github/workflows/rlrs-ci.yml), Jira (project RLRS).
Test Data	rlrs_app/seed.py: 4 synthetic users (student1, student2, incharge, admin @lab.test), 2 labs (Electronics, Fabrication), 3 equipment items (Digital Oscilloscope, Logic Analyzer, 3D Printer requiring approval).

7. Test Schedule

Activity	Planned Window	Actual Execution & Outcome
Review SRS/SAD & prepare test cases	Sprint 1 (27–31 Oct 2026)	Completed: 29 baseline + 4 security/health test cases defined in STP.
Unit tests (pure scheduling engine)	Sprint 1 (27–31 Oct 2026)	Completed 31 Oct: 37 unit tests passing (docs/test_evidence_unit_2026-10-31.txt).

Activity	Planned Window	Actual Execution & Outcome
Integration, API, concurrency & security tests	Sprint 2 (2–7 Nov 2026)	Completed 5–11 Nov: 71 automated pytest tests passing across Linux, Windows, and macOS.
Browser end-to-end smoke test	Sprint 2 (2–7 Nov 2026)	Completed 5 Nov: 9/9 Playwright Chromium steps passing with screenshots in tests/ui/screenshots/.
Final acceptance run & demo readiness	By 11 Nov 2026	Completed: All 15 Jira stories verified and marked Done; v1.1 documents aligned.

8. Test Deliverables & 9. Roles and Responsibilities

Role	Assigned Team Member	Responsibilities
Test Coordinator / QA, Notifications, CI & Docs	Mahipal Singh (PES1UG24AM153)	Maintain STP and RTM; implement and verify RLRS-1, 10, 12, 13, 14, 15 (reminders, maintenance, attendance, audit, reports, CI, v1.1 alignment).
Scheduling Engine & Backend Developer	Lakshya Budhauliya (PES1UG24AM147)	Implement and verify RLRS-4, 5, 6, 11 (conflict detection, booking rules, concurrency-safe UTC reservation API, cross-lab approval checks).
Calendar UI, Auth, Equipment & Security Developer	Deepesh M (PES1UG24AM150)	Implement and verify RLRS-2, 3, 7, 8, 9 (sign-in/RBAC, lab/equipment management, weekly calendar UI, cancellation, holder-only remote_info, macOS verification).
Documentation & Verification Support	Venkat Kushal (PES1UG24AM124)	Assist with test plan review, document formatting verification, and cross-platform walkthrough.

10. Risks & Mitigation, 11. Assumptions & 12. Suspension Criteria

- Risks & Mitigation: Concurrency race conditions mitigated via SQLite BEGIN IMMEDIATE lock + 10-thread barrier test; flaky timer tests avoided via injected clock; accidental emails prevented via FileTransport.
- Assumptions: Synthetic .test emails only; local SQLite file per test via pytest tmp_path fixture.
- Suspension Criteria: Suspend testing if server fails /api/health check or if any test writes outside temporary directories.

13. Test Case Management and Traceability

Test Case ID	Scenario	Expected Result	Requirement(s)	Automated Test	Result
TC-AUTH-01	Sign in with valid/invalid credentials & session lifecycle	200 + session cookie on valid; 401 on bad password or after logout	RLRS-F-001, RLRS-SR-001	test_requires_login_and_session_lifecycle, test_register_login_and_bad_credentials	Pass
TC-AUTH-02	Student calls staff endpoints directly	403 Forbidden; no database state change	RLRS-F-002, RLRS-SR-002	test_role_enforcement_is_server_side	Pass
TC-EQ-01	Lab in-charge creates, updates, and searches equipment	Equipment saved and returned by search filter	RLRS-F-003, RLRS-F-005	test_equipment_management_permissions_and_search	Pass
TC-EQ-02	Student attempts to book deactivated equipment	409 EQUIPMENT_INACTIVE; no reservation created	RLRS-F-003	test_inactive_equipment_cannot_be_booked	Pass
TC-CAL-01	Open week view for equipment with bookings & maintenance	Slots show Free, Yours, Booked, Maintenance with text labels	RLRS-F-004, RLRS-F-006	test_availability_marks_own_booked_and_validates_range, test_lab_hours_change_drives_the_calendar_grid	Pass
TC-BOOK-01	Book a free 10:00–11:00 slot	201 Confirmed; UTC stored in DB; confirmation email queued	RLRS-F-007, RLRS-F-014	test_booking_creates_reservation_email_reminder_audit, test_rlrs6_timestamps_stored_in_utc	Pass
TC-CONF-01	Book 10:30–11:30 when 10:00–11:00 is booked	409 CONFLICT with conflicting time range; transaction rolled back	RLRS-F-008	test_partial_overlap_rejected_with_range, test_conflict_returns_409_with_range_and_rolls_back	Pass
TC-CONF-02	Book 11:00–12:00 when 10:00–11:00 is booked	201 Accepted (back-to-back half-open intervals do not conflict)	RLRS-F-008	test_touching_back_to_back_is_accepted	Pass
TC-CONF-03	Book 09:00–12:00 containing an existing 10:00–11:00 booking	409 CONFLICT; containment detected	RLRS-F-008	test_containing_and_contained_rejected	Pass
TC-CONF-04	Book a slot after previous holder cancelled it	201 Accepted	RLRS-F-008, RLRS-F-012	test_cancelled_slot_can_be_rebooked	Pass
TC-RULE-01	Book in the past or outside lab operating hours	400 RULE_VIOLATION (PAST_START, OUTSIDE_HOURS)	RLRS-F-009	test_past_start, test_outside_hours_boundaries	Pass
TC-RULE-02	Book beyond advance window or above max duration	400 RULE_VIOLATION (BEYOND_ADVANCE_WINDOW, EXCEEDS_MAX_DURATION)	RLRS-F-009	test_advance_window_edge, test_max_duration_edge	Pass
TC-RULE-03	Student at max active bookings requests one more	400 RULE_VIOLATION (MAX_ACTIVE_EXCEEDED)	RLRS-F-009	test_max_active_per_user	Pass
TC-RULE-04	Student books two different equipment items at overlapping times	400 RULE_VIOLATION (USER_OVERLAP)	RLRS-F-009	test_user_cannot_double_book_same_time_on_other_equipment	Pass

Test Case ID	Scenario	Expected Result	Requirement(s)	Automated Test	Result
TC-APPR-01	Book approval-required equipment; approve/reject & cross-lab check	Pending on create; other lab in-charge gets 403; assigned in-charge approves/rejects	RLRS-F-010, RLRS-SR-005	test_approval_flow_and_rejection, test_rlrs11_other_lab_incharge_denied	Pass
TC-CANC-01	Cancel own reservation >60 min before start	Status Cancelled; slot freed; pending reminder cancelled	RLRS-F-012	test_cancel_frees_slot_and_cancels_reminder	Pass
TC-CANC-02	Student cancels own reservation <60 min before start	409 CUTOFF_PASSED; reservation remains Confirmed	RLRS-F-012	test_cancel_after_cutoff_refused	Pass
TC-MAINT-01	Create maintenance block overlapping active reservation	Overlapping booking Cancelled; holder emailed; new bookings blocked	RLRS-F-013	test_maintenance_cancels_overlapping_and_blocks_new_bookings	Pass
TC-MAIL-01	SMTP failure when sending queued email/reminder	Retried with 5-min backoff up to 3 attempts, then marked Failed	RLRS-F-016	test_emails_flushed_once_and_retried_on_failure, test_repeated_failure_marks_failed	Pass
TC-MAIL-02	Advance fake clock to 2h before start and run worker twice	Exactly one reminder sent	RLRS-F-015, RLRS-NF-004	test_sent_exactly_once_even_if_worker_runs_twice	Pass
TC-MAIL-03	Cancel reservation then advance clock past reminder time	Zero reminders sent	RLRS-F-015	test_no_reminder_after_cancellation	Pass
TC-ATT-01	Mark attendance before and after slot end	409 before end; Completed/NoShow recorded after end	RLRS-F-018	test_attendance_only_after_slot_end_and_audited	Pass
TC-REP-01	Generate utilisation report for known dataset	Booked hours, available hours, % and no-show counts match exactly	RLRS-F-019	test_utilisation_report_reconciles, test_known_dataset_reconciles	Pass
TC-AUDIT-01	Inspect audit records after login and booking changes	Actor, action, timestamp recorded; secrets redacted	RLRS-F-021, RLRS-SR-007	test_audit_records_actor_action_time_and_redacts_secrets	Pass
TC-CONC-01	10 threads simultaneously book the exact same slot	Exactly 1 succeeds (201); 9 receive 409 CONFLICT	RLRS-NF-002	test_concurrent_requests_for_same_slot_yield_exactly_one_booking	Pass
TC-PERF-01	20+ requests on 200-equipment / 20k-reservation dataset	90% complete within 2 seconds	RLRS-NF-001	none (bulk dataset not built)	Not run
TC-RESTART-01	Simulated server restart with pending reminders in SQLite	Data intact; each reminder sent once after restart	RLRS-NF-003	test_reminder_sent_two_hours_before_exactly_once_and_survives_restart	Pass
TC-SEC-01	Inspect stored password hashes, cookies, logs & repo	PBKDF2 salted hashes; HttpOnly cookie; zero plaintext secrets	RLRS-NF-005, SR-003, SR-006	test_audit_never_contains_password, test_cookie_is_httponly	Pass
TC-SEC-02	Student cancels another's booking or reads unconfirmed remote_info	403 on cancel; remote_info empty unless Confirmed holder or staff	RLRS-SR-005, RLRS-F-017	test_cancel_over_http_and_other_user_forbidden, test_rlrs8_remote_info_restricted_to_confirmed_holder_and_staff	Pass

Test Case ID	Scenario	Expected Result	Requirement(s)	Automated Test	Result
TC-HEALTH-01	GET /api/health without authentication	200 OK with {status: ok, db: ok}	RLRS-NF-008	test_health_endpoint_is_public_and_checks_db	Pass
TC-SEC-03	SQL injection payloads in login and equipment search	Treated as literal strings; 401/empty list; tables intact	RLRS-SR-004	test_sql_injection_strings_are_inert	Pass
TC-SEC-04	Static path traversal requests (/../service.py)	404 Not Found; no files outside static/ served	RLRS-SR-004	test_static_page_served_and_path_traversal_blocked	Pass
TC-ERR-01	Trigger malformed JSON, bad IDs, and 404 endpoints	Structured JSON error returned; zero Python stack traces leaked	RLRS-F-022	test_errors_are_json_without_stack_traces	Pass

14. Test Metrics, Execution Summary and Reporting

Final Execution Summary (as of 11 November 2026): 71 automated pytest tests executed, 71 passed, 0 failed (37 unit tests + 34 integration/API tests), verified across Ubuntu Linux, Windows, and macOS in GitHub Actions CI as well as locally on macOS (Darwin Python 3.14.7, docs/test_evidence_2026-11-11_macos_deepesh.txt). Browser end-to-end smoke test: **9/9 steps passed** (Playwright Chromium). Total planned test cases: 33 (29 baseline + 4 added); executed: 32; **passed: 32 (100% of executed)**; partial: 0; not run: 1 (TC-PERF-01 bulk dataset). Double-booking incidents: 0. Duplicate or missed reminders: 0. Open security defects: 0.

15. Approvals

Role	Name	Signature / Status
Test Coordinator / QA Lead	Mahipal Singh (PES1UG24AM153)	Approved — 11 Nov 2026
Scheduling Engine Developer	Lakshya Budhauliya (PES1UG24AM147)	Approved — 11 Nov 2026
Calendar UI & Security Developer	Deepesh M (PES1UG24AM150)	Approved — 11 Nov 2026
QA & Documentation Member	Venkat Kushal (PES1UG24AM124)	Approved — 11 Nov 2026
Course Instructor / TA	Kshitij Shetty	Pending Evaluation